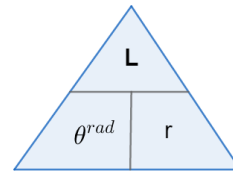


$$\begin{aligned}\sin(180 - \theta) &= \sin \theta & \csc(180 - \theta) &= \csc \theta \\ \cos(180 - \theta) &= -\cos \theta & \sec(180 - \theta) &= -\sec \theta \\ \tan(180 - \theta) &= -\tan \theta & \cot(180 - \theta) &= -\cot \theta\end{aligned}$$



S	A
(180-θ)	(360+θ)
(90+θ)	(90-θ)
(180+θ)	(360-θ)
(270-θ)	(270+θ)
T	C

$$\begin{aligned}\sin(180 + \theta) &= -\sin \theta & \csc(180 + \theta) &= -\csc \theta \\ \cos(180 + \theta) &= -\cos \theta & \sec(180 + \theta) &= -\sec \theta \\ \tan(180 + \theta) &= \tan \theta & \cot(180 + \theta) &= \cot \theta\end{aligned}$$

$$\frac{x^\circ}{180} = \frac{\theta^{rad}}{\pi} = \frac{y^{grad}}{200}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\begin{aligned}\sin(360 - \theta) &= -\sin \theta & \csc(360 - \theta) &= -\csc \theta \\ \cos(360 - \theta) &= \cos \theta & \sec(360 - \theta) &= \sec \theta \\ \tan(360 - \theta) &= -\tan \theta & \cot(360 - \theta) &= -\cot \theta\end{aligned}$$

$$\cosh^2 \theta - \sinh^2 \theta = 1 \quad 1 - \tanh^2 \theta = \operatorname{sech}^2 \theta$$

$$\coth^2 \theta - 1 = \operatorname{csch}^2 \theta$$

$$\begin{aligned}\sin(-\theta) &= -\sin \theta & \csc(-\theta) &= -\csc \theta \\ \cos(-\theta) &= \cos \theta & \sec(-\theta) &= \sec \theta \\ \tan(-\theta) &= -\tan \theta & \cot(-\theta) &= -\cot \theta\end{aligned}$$

$$\sinh 2\theta = 2 \sinh \theta \cosh \theta$$

$$\cosh 2\theta = \cosh^2 \theta + \sinh^2 \theta = 1 + 2 \sinh^2 \theta = 2 \cosh^2 \theta - 1$$

$$\tanh 2\theta = \frac{2 \tanh \theta}{1 - \tanh^2 \theta}$$

$$\begin{aligned}\sin(90 - \theta) &= \cos \theta & \csc(90 - \theta) &= \sec \theta \\ \cos(90 - \theta) &= \sin \theta & \sec(90 - \theta) &= \csc \theta \\ \tan(90 - \theta) &= \cot \theta & \cot(90 - \theta) &= -\tan \theta\end{aligned}$$

$$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$$

$$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$$

$$\tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$$

$$\begin{aligned}\sin(90 + \theta) &= \cos \theta & \csc(90 + \theta) &= \sec \theta \\ \cos(90 + \theta) &= -\sin \theta & \sec(90 + \theta) &= -\csc \theta \\ \tan(90 + \theta) &= -\cot \theta & \cot(90 + \theta) &= -\tan \theta\end{aligned}$$

$$\sin(270 - \theta) = -\cos \theta \quad \csc(270 - \theta) = -\sec \theta$$

$$\cos(270 - \theta) = -\sin \theta \quad \sec(270 - \theta) = -\csc \theta$$

$$\tan(270 - \theta) = \cot \theta \quad \cot(270 - \theta) = \tan \theta$$

In any tri. fun. Equ. then divide by ($\sin x$, $\cos x$,)

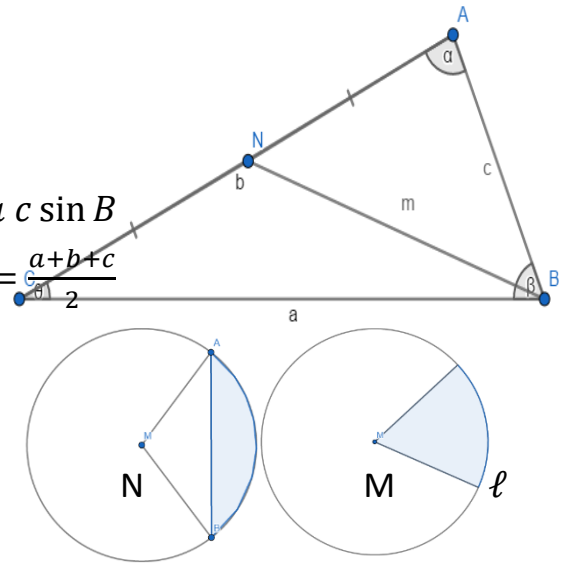
$$\sin(270 + \theta) = -\cos \theta \quad \csc(270 + \theta) = -\sec \theta$$

$$\cos(270 + \theta) = \sin \theta \quad \sec(270 + \theta) = \csc \theta$$

$$\tan(270 + \theta) = -\cot \theta \quad \cot(270 + \theta) = -\tan \theta$$

❖ In ΔABC :

- Area of $\Delta ABC = \frac{1}{2} B.H. = \frac{1}{2} \|\vec{A} \times \vec{B}\|$
 $= \frac{1}{2} ab \sin C = \frac{1}{2} b c \sin A = \frac{1}{2} a c \sin B$
 $= \sqrt{s(s-a)(s-b)(s-c)}$, Where: $s = \frac{a+b+c}{2}$
 $= \frac{abc}{4r}$, (r = radius of circumcircle)
- $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \frac{a+b+c}{\sin A + \sin B + \sin C} = 2r$
- $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$
- $\sin A \sin B \sin C = \frac{abc}{8r^3}$
- $r^2 = \frac{abc}{4 \sin A \sin B \sin C}$
- $m = BN = \frac{1}{2} \sqrt{2a^2 + 2c^2 - b^2}$



❖ In circle M:

- Area of shaded part $= \frac{\theta^{\text{rad}} r^2}{2} = \frac{x^\circ \pi r^2}{360} = \frac{\ell r}{2}$

❖ In circle N:

- Area of shaded part $= \frac{r^2 (\theta^{\text{rad}} - \sin \theta^\circ)}{2}$

❖ In $\Delta 1$:

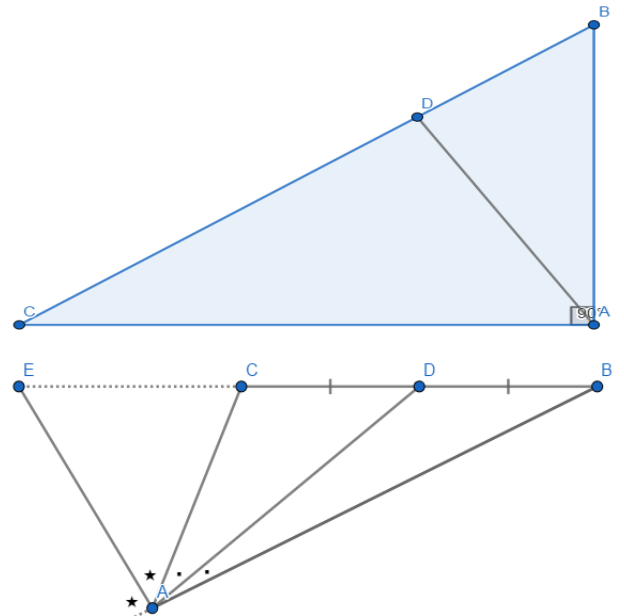
- $(AB)^2 = BD \cdot BC$
- $(AC)^2 = CD \cdot CB$
- $(AD)^2 = BD \cdot CD$
- $AD = \frac{AB \cdot AC}{BC}$

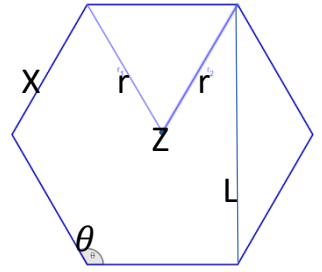
❖ In $\Delta 2$:

- $AD = \sqrt{AB \cdot AC - BD \cdot DC}$
- $AE = \sqrt{BE \cdot EC - AB \cdot AC}$
- $AB \cdot AC = AD \cdot BC$

❖ Similarity:

If $ABC \sim XYZ \therefore \frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ} = \frac{P_1}{P_2} = \frac{h_1}{h_2} = \frac{\sqrt{A_1}}{\sqrt{A_2}}$





❖ Area of poly. $Z = \frac{n}{4} x^2 \cot \frac{\pi}{n} = \frac{n}{2} r^2 \sin \frac{2\pi}{n}$

❖ $m(\angle \theta) = \frac{(n-2) \times 180}{n}$ $L = X\sqrt{3}$ $r=X$

❖ Cylinder :

• $\text{Vol} = \pi r^2 h$, $\text{L.A} = 2\pi r h$, $\text{T.S} = 2\pi r (h + r)$

❖ cone :

• $V_{ol} = \frac{1}{3} \pi r^2 . h$, $\text{L.A} = \pi r L$, $\text{T.A} = L . A + \pi r^2$

❖ Sphere : $V_{ol} = \frac{4}{3} \pi r^3$, $\text{T.A} = 4\pi r^2$

❖ Pyramid:

• $\text{L.A} = \frac{1}{2} \text{base perimeter} \times h'$, $\text{T.A} = \text{L.A} + \text{area of base}$, $V_{ol} = \frac{1}{3} \times \text{area of base} \times h$

• In Tri. Pyramid of regular faces: $3h^2 = 2L^2$, $\text{T.A} = L^2\sqrt{3}$, $V_{ol} = \frac{\sqrt{2}L^3}{12}$

❖ Area of rectangle $= \frac{1}{2} d_1 . d_2 \sin \theta$

❖ If $A = (x_1, y_1)$ $B = (x_2, y_2) \therefore A + B = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

❖ The eq. of straight line :

1- Given point (x_1, y_1) and slop (m) is: $y - y_1 = m (x - x_1)$

2- Given slope (m) and intarcepted part from y-axis is : $y = mx + c$

3- Given intarcepted parts for the two coordinate axes is: $\frac{x}{a} + \frac{y}{b} = 1$

❖ Rules of derivative:

Rule 1: $y = a \therefore y' = \text{Zero}$ **Rule 2:** $y = ax \therefore y' = a$ **Rule 3:** $y = x^n \therefore y' = nx^{n-1}$

Rule 4: $y = f(x) \cdot g(x) \therefore y' = f(x) \cdot g'(x) + g(x) \cdot f'(x)$ **Rule 5:** $y = \frac{f(x)}{g(x)} \therefore y' = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{g^2(x)}$

Rule 6: $y = f^n(x) \therefore y' = n f^{n-1}(x) \cdot f'(x)$ **Rule 7:** $y = e^x \therefore y' = e^x$

Rule 8: $y = a^x \therefore y' = a^x \ln a$ **Rule 9:** $y = \ln x \therefore y' = \frac{1}{x}$ **Rule 10:** $y = \log_a x \therefore y' = \frac{1}{x} \log_a e$

Rule 11: $y = \sin x \therefore y' = \cos x$ **Rule 12:** $y = \cos x \therefore y' = -\sin x$ **Rule 13:** $y = \tan x \therefore y' = \sec^2 x$

Rule 14: $y = \csc x \therefore y' = -\csc x \cdot \cot x$ **Rule 15:** $y = \sec x \therefore y' = \sec x \cdot \tan x$

Rule 16: $y = \cot x \therefore y' = -\csc^2 x$ **Rule 17:** $y = (f \circ g)(x) \therefore y' = f'(g(x)) \cdot g'(x)$

Rule 18: $y = \sin ax \therefore \frac{d^n y}{dx^n} = a^n \sin \left(ax + \frac{n\pi}{2} \right)$ **Rule 19:** $y = \ln x \therefore \frac{d^n y}{dx^n} = (-1)^{n-1} (n-1)! \cdot x^{-n}$

F	F'
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\text{sech}^2 x$
$\coth x$	$-\text{csch}^2 x$
$\text{sech } x$	$-\text{sech } x \tanh x$
$\text{csch } x$	$-\text{csch } x \coth x$

$\tan^{-1} x$ $\frac{\pm 1}{1+x^2}$ $-\cot^{-1} x$	$\tanh^{-1} x$ $\frac{1}{1-x^2}$	$-\coth^{-1} x$ $\frac{1}{x^2-1}$
$\sin^{-1} x$ $\frac{\pm 1}{\sqrt{1-x^2}}$ $-\cos^{-1} x$	$\sinh^{-1} x$ $\frac{1}{\sqrt{1+x^2}}$	$\cosh^{-1} x$ $\frac{1}{\sqrt{x^2-1}}$
$\sec^{-1} x$ $\frac{\pm 1}{x\sqrt{x^2-1}}$ $-\csc^{-1} x$	$-\text{sech}^{-1} x$ $\frac{1}{x\sqrt{1-x^2}}$	$-\text{csch}^{-1} x$ $\frac{1}{x\sqrt{1+x^2}}$

❖ Rules of integration:

Rule 1: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Rule 4: $\int x^n \cdot x' dx = \frac{x^{n+1}}{n+1} + C$

Rule 7: $\int 6x^2 a^{2x^3} dx = \frac{a^{2x^3}}{\ln a} + C$

Rule 10: $\int \cos x dx = \sin x + C$

Rule 11: $\int \tan x dx = \ln |\sec x| + C = -\ln |\cos x| + C$

Rule 12: $\int \cot x dx = \ln |\sin x| + C = -\ln |\csc x| + C$

Rule 13: $\int \sec x dx = \ln |\sec x + \tan x| + C$

Rule 14: $\int \csc x dx = -\ln |\csc x + \cot x| + C$

Rule 15: $\int \sec^2 x dx = \tan x + C$

Rule 16: $\int \csc^2 x dx = -\cot x + C$

Rule 17: $\int \sin^2 x dx = \frac{x}{2} - \frac{1}{4} \sin 2x + C$

Rule 18: $\int \cos^2 x dx = \frac{x}{2} + \frac{1}{4} \sin 2x + C$

Rule 19: $\int \tan^2 x dx = \tan x - x + C$

Rule 20: $\int \cot^2 x dx = -\cot x - x + C$

To remove "Ln": $e^{\ln a} = a$

Rule 2: $\int (ax + b)^n dx = \frac{1}{a} \frac{(ax+b)^{n+1}}{n+1} + C$

Rule 5: $\int a^x dx = \frac{a^x}{\ln a} + C$

Rule 8: $\int \frac{x'}{x} dx = \ln |x| + C$

Rule 3: $\int \frac{1}{x} dx = \ln |x| + C$

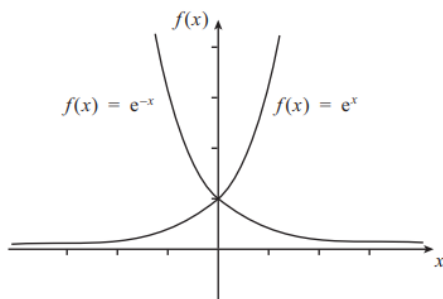
Rule 6: $\int e^x dx = e^x + C$

Rule 9: $\int \sin x dx = -\cos x + C$

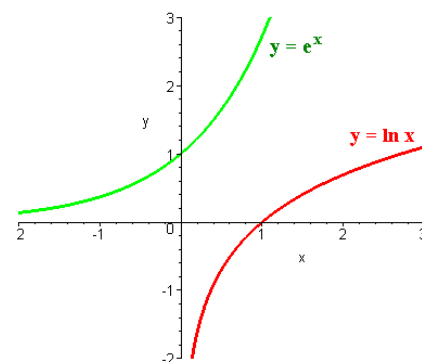
$\lim_{x \rightarrow \infty} \left(1 - \frac{a}{x}\right)^{n+k} = e^{-a}$	$\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^{\frac{x}{a}} = e$
$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{ax} = e^a$	$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{x+a} = e$
$\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^a$	$\lim_{x \rightarrow 0} e^x = 1$
$\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x}\right) = 1$	$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
$\lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x}\right) = \ln a$	$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$
$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = \log e$	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
$\lim_{x \rightarrow \infty} (1+x)^{\frac{a}{x}} = e^a$	$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$
$\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m} a^{n-m}$	$\lim_{x \rightarrow 0} \frac{(x+a)^n - a^n}{x} = na^{n-1}$
$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \text{Zero}$	

❖ *Exponential function:*

$$y = e^x$$

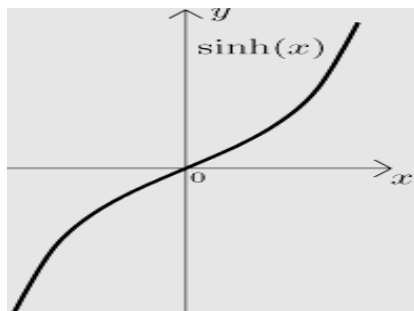


$$y = \ln x$$

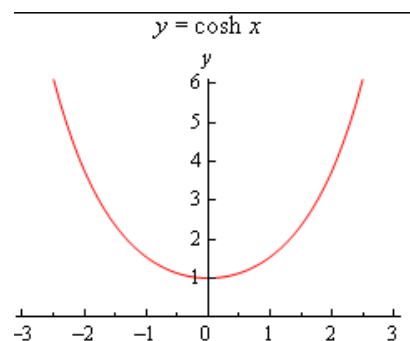


❖ *Hyperbolic function:*

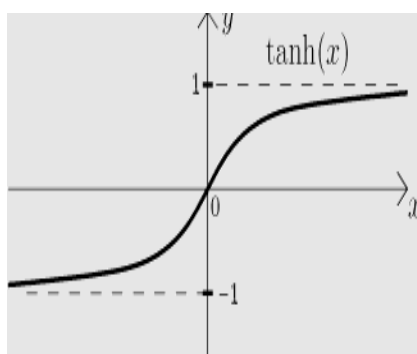
$$y = \sinh x$$



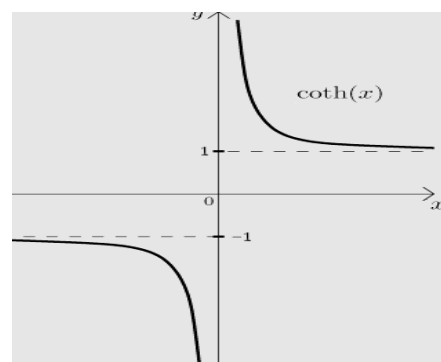
$$y = \cosh x$$



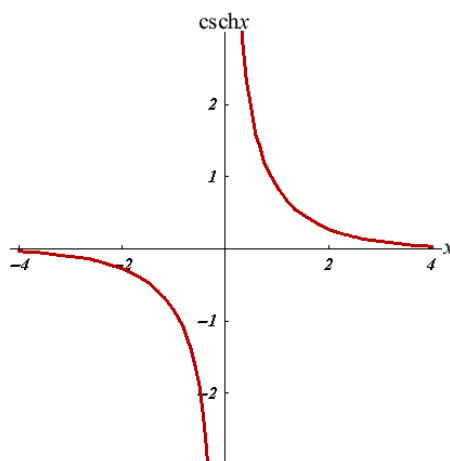
$$y = \tanh x$$



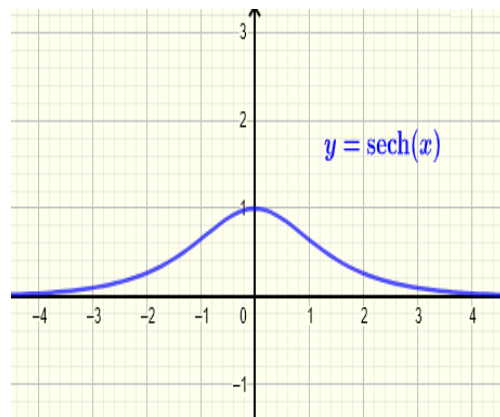
$$y = \coth x$$



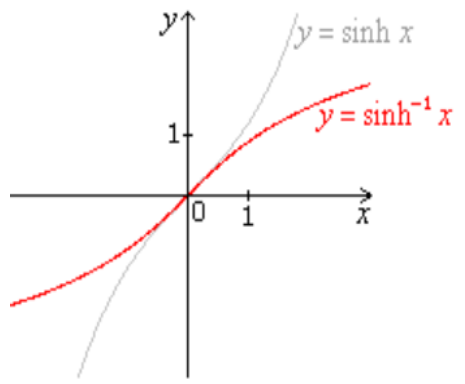
$$y = \operatorname{csch} x$$



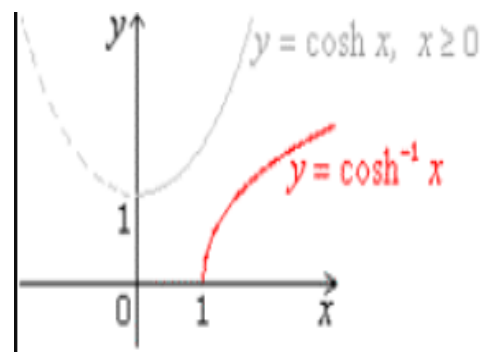
$$y = \operatorname{sech} x$$



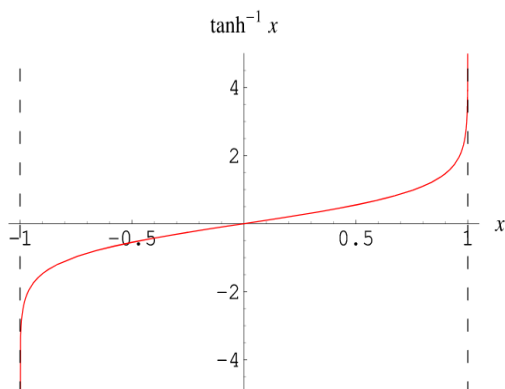
$$y = \sinh^{-1} x$$



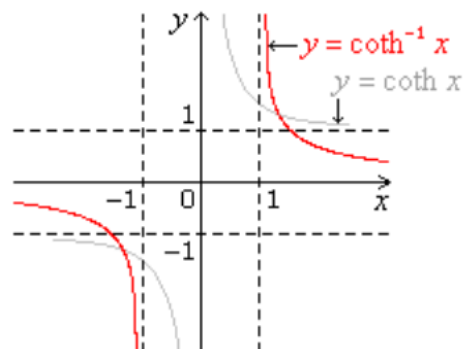
$$y = \cosh^{-1} x$$



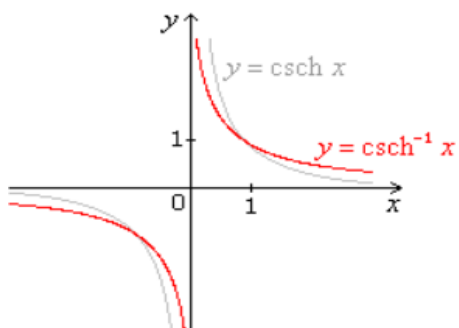
$$y = \tanh^{-1} x$$



$$y = \coth^{-1} x$$



$$y = \operatorname{csch}^{-1} x$$



$$y = \operatorname{sech}^{-1} x$$

